

Kotaro Ikeda

📍 Tokyo, Japan ✉ kotatumuri114@icloud.com 🌐 kotatumuri-room.github.io

About

My name is Kotaro Ikeda. I am interested in machine learning as well as its intersections with thermodynamics and dynamical systems.

Education

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|-----------|---|---------------------|
| | The University of Tokyo , Junior Division, College of Arts and Sciences – Tokyo, Japan | Apr 2022 – Mar 2024 |
| BE | The University of Tokyo , Department of Mathematical Engineering and Information Physics, School of Engineering – Tokyo, Japan | Apr 2024 – present |
| | • Planned to graduate in March 2026 | |

Publications

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|--|-----------|
| Speed-accuracy relations for diffusion models: Wisdom from nonequilibrium thermodynamics and optimal transport | June 2025 |
| <i>Kotaro Ikeda</i> , Tomoya Uda, Daisuke Okanohara, Sosuke Ito
10.1103/x5vj-8jq9 (Phys. Rev. X) | |
| Pairwise Optimal Transports for Training All-to-All Flow-Based Condition Transfer Model | Dec 2025 |
| <i>Kotaro Ikeda</i> , Masanori Koyama, Jinzhe Zhang, Kohei Hayashi, Kenji Fukumizu
10.48550/arXiv.2504.03188 (NeurIPS 2025) | |

Projects

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|---|---------------------|
| Research Experience Seminar on the Universal Principles of Life | Apr 2023 – Mar 2024 |
| Small-group seminar by Universal Biology Institute, the University of Tokyo | |
| <ul style="list-style-type: none">• Constructed a novel analogy between diffusion models and stochastic thermodynamics• Theoretically discussed the optimality of noise schedules in diffusion models• Published a paper on this topic in a top-physics journal• Press Release is here | |
| Board Game agent development@May Fes | Feb 2025 – May 2025 |
| Created a reinforcement learning based agent for Blokus™ | |
| <ul style="list-style-type: none">• State-of-the-art performance against publically available agents.• Created a website and a real-world competition system for playing with our agents.• Project done for May Fes with my classmates.• Served as a project leader. | |

Fundings

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|--|----------------------|
| Funai Overseas Scholarship | Sept 2026 – Aug 2028 |
| Partial funding for studying abroad. | |
| <ul style="list-style-type: none">• \$36,000 per year for living cost• \$14,000 per year for tuition• full coverage of medical insurance | |

Experience

Summer Intern, Preferred Networks, Inc. – Tokyo, Japan

Aug 2024 – Feb 2025

- Developed deep learning models for style transfer based on flow matching and optimal transport theories.
- Preprint of this work can be found in the publications section.

Part-time Engineer, Akari, Inc. – Tokyo, Japan

June 2023 – Mar 2024

- Enrolled in a project involving NeRF and 3D Gaussian splatting.

Part-time Engineer, Meltly, Inc. – Tokyo, Japan

Apr 2025 – present

- Worked for training a generative model for anime-character images.

Skills

Programming: Proficient with Python, PyTorch and Git; good understanding of C++, Web development (React) and App development (Swift, Java)

Languages: English (fluent, TOEFL: 105/120), Japanese (native)

Conferences in Japan

Speed-accuracy relations for diffusion models: Wisdom from nonequilibrium thermodynamics and optimal transport

Nov 2024

Kotaro Ikeda, Tomoya Uda, Daisuke Okanohara, Sosuke Ito

The 27th Information-Biased Induction Sciences Workshop (IBIS), Saitama, Japan

GeoBeta Flow Matching

Nov 2024

Kotaro Ikeda, Masanori Koyama, Jinzhe Zhang, Kohei Hayashi, Kenji Fukumizu

The 27th Information-Biased Induction Sciences Workshop (IBIS), Saitama, Japan